Assignment

Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend development focuses on the visible parts of a website that users interact with directly, using HTML, CSS, and JavaScript.

Backend development handles server-side logic, databases, and data processing using technologies like Node.js, Python, or PHP.

Full-stack development involves both frontend and backend development.

Example :

When you visit GLA portal, the Dashboard, Mapped Classes etc. are part of frontend, the login credentials' (roll no., password) is verified by backend, and the combined development of both comes under full-stack.

Create a simple diagram showing how the client-server model works in web architecture.

```
[ Client / Browser ]
        |
  (HTTP Request)
        ↓
[ Web Server / Backend ]
        |
 (Database Query)
        ↓
[ Database Server ]
        ↑
 (Response Data)
        |
 (HTTP Response)
        ↑
[ Client displays web page ]
```

Describe how a browser requests and displays a web page from a web server.

1. User enters a URL in the browser.
2. Browser sends a DNS request to find the server's IP address.
3. A HTTP/HTTPS request is sent to the web server.
4. Server processes the request and sends back an HTML response.
5. Browser parses the HTML and requests CSS, JS, and images.
6. Rendering engine displays the page visually.

Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Tool	Purpose
VS Code	Code editor for writing HTML, CSS, and JS
Web Browser	To test and preview web pages
Node.js & npm	For running JS and managing packages
Git & GitHub	For version control and collaboration
Live Server Extension	Auto-refresh browser during development
XAMPP/WAMP	Local server for PHP/MySQL

Explain what a web server is and give examples of commonly used servers.

A web server is software that stores, processes, and delivers web pages to clients upon request. Examples include Apache, Nginx, Microsoft IIS, LiteSpeed, and Node.js (Express.js).

Define the roles of a frontend developer, backend developer, and database administrator in a project.

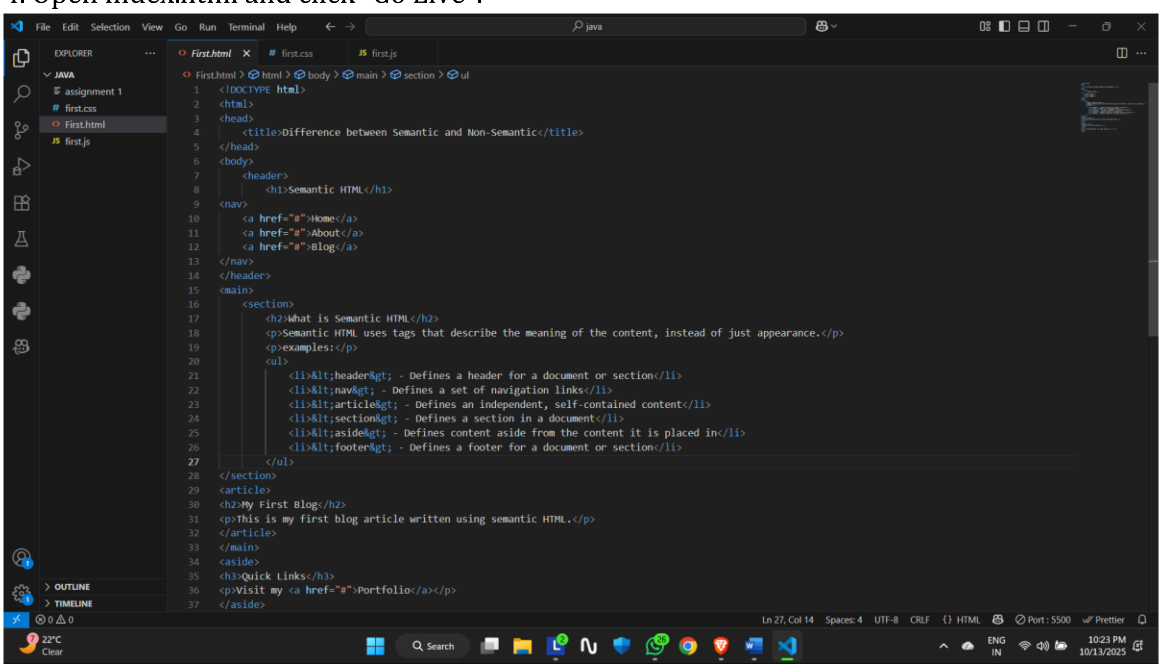
Role	Responsibilities
Frontend Developer	Builds the user interface using HTML, CSS, and JavaScript.

Backend Developer	Handles data, authentication, and server logic.
Database Administrator	Maintains and secures the database, ensuring performance.

Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Steps:

1. Download VS Code from https://code.visualstudio.com/
2. Install extensions:
   - Live Server
   - HTML CSS Support
   - Prettier
   - JavaScript (ES6) Snippets
3. Create a folder and add index.html, style.css, script.js.
4. Open index.html and click “Go Live”.



Explain the difference between static and dynamic websites. Provide an example of each.

Type	Description	Example
Static	Displays fixed content, doesn’t change unless updated manually.	HTML/CSS company site
Dynamic	Changes content dynamically using database and server logic.	Facebook, YouTube

Research and list five web browsers. Explain how rendering engines differ between them.

Browser	Rendering Engine	Description
Google Chrome	Blink	Fast, supports modern standards.
Mozilla Firefox	Gecko	Open-source, privacy focused.
Safari	WebKit	Optimized for Apple devices.
Microsoft Edge	Blink	Chromium-based, good Windows integration.
Opera	Blink	Lightweight with VPN and ad-blocker.

Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

```
[ Client (UI) ]
  |
  ↓
[ Web Server / API ]
  |
  ↓
[ Database Server ]
  |
  ↓
[ Response sent back to client ]
```